30 Solve the simultaneous equations

$$3x - 4y = 11$$
$$9x + 2y = 5$$

(1) $3x - 4y = 11$

(2) $9x + 2y = 5$

$2 \times (2)$: $18x + 4y = 10$

Opposite y coefficients: add (1) and $2 \times (2)$.

$21x = 21$

$\div 21 \quad \div 21$

$x = 1$

1 mark
Eliminate y by adding.

Substitute $x = 1$ in (2):

$9 + 2y = 5$

$-9 \quad -9$

$2y = -4$

$\div 2 \quad \div 2$

$y = -2$

1 mark
Substitute into an original equation.

1 mark
x = 1 and y = −2.

$x =$ 1

$y =$ −2

(Total for Question 30 is 3 marks)